

DEWA

Dubai Electricity & Water Authority

Office of the CAIO · AI-Native Enterprise Programme

AI-Native Enterprise & Digital Workforce Framework

Organization-Led Transformation, Human-Agent Operating Model & Harness Engineering for Quality, Value and Token Economics

A practical enterprise framework that starts from the organization structure, maps human and agent work to quality procedures and processes, governs agents through engineered harnesses, and measures strategic outcomes, excellence, value and AI consumption.

19

Framework Sections & Appendices

10

Connected Framework Pillars

6-Stage

AI-Native Maturity Progression

CLASSIFICATION Internal — CAIO / AI Governance

PREPARED FOR Chief AI Officer (CAIO) Direction

ENTERPRISE CONTEXT DEWA — AI-native operating model

RECOMMENDED IMPLEMENTATION SAP Neptune-led portal with federated AI, analytics and observability

VERSION Consolidated Framework — July 2026

Contents

PART A VISION & FRAMEWORK FOUNDATION

- 1** Executive Summary
- 2** CAIO Vision and Framework Principles
- 3** Enterprise Framework Overview

PART B HUMAN-AGENT ORGANIZATION & PROCESS DESIGN

- 4** Human-Agent Organization and Digital Workforce
- 5** Quality Procedures and Process Agenticity

PART C HARNESS ENGINEERING & AI PLAYBOOK

- 6** Harness Engineering
- 7** AI Transformation Playbook

PART D DATA, VALUE & TOKEN ECONOMICS

- 8** Data, Integration and Measurement Model
- 9** Value Realization and Token Economics

PART E TECHNOLOGY & GOVERNANCE

- 10** Technology and Visualization Architecture
- 11** Governance, Risk and Performance

PART F ROADMAP & EXECUTION

- 12** Implementation Roadmap
- 13** Recommended MVP and D2D Pilot
- 14** Executive Dashboards and KPIs

PART G DECISIONS & APPENDICES

- 15** Key Decisions and Success Factors
- A** Appendix A · Reference Data Model
- B** Appendix B · Illustrative Agent and Harness Records
- C** Appendix C · CAIO Whiteboard Reference
- D** Appendix D · External Reference Sources

PART A

Vision & Framework Foundation

Sections 1–3

1. Executive Summary

The proposed framework establishes an AI-native enterprise operating model in which organization units, human roles, AI copilots, managed agents and autonomous digital workers are represented in one governed structure. It moves AI transformation away from isolated use cases and connects every initiative to strategic objectives, excellence criteria, section mandates, Quality Procedures, jobs, activities, processes, controls, outcomes, value and cost.

Core transformation chain

Organization → Section Mandate → Quality Procedure → Process and Activity → Human/Agent Allocation → Harness → Operational Evidence → Business Outcome → Value Realization → Token Economics

The recommended solution is a hybrid architecture. SAP Neptune should provide the unified business portal, workflow and master-data maintenance experience. A custom interactive graph should provide the organization and human-agent visualization. D2D should provide work-item and outcome evidence. Microsoft Copilot reports and APIs should provide adoption and usage indicators. Apache Superset should provide executive and Value Realization analytics, while Grafana should provide agent and harness observability. A shared enterprise data and control layer should prevent dependency on any single agent platform or visualization tool.

Harness Engineering is a formal pillar of this framework. The harness is the engineered operating environment around the model or agent: instructions, tools, context, workflow, permissions, guardrails, evaluations, logging, human checkpoints, token budgets, fallbacks and lifecycle controls. Quality Procedures are a primary source for designing the harness and for verifying that an agent follows the approved way of working.

Executive question	Framework response
Where are AI and agents working?	Human-agent organization map from division to section, position, process and agent.
How much work is covered by AI?	Activity-level AI contribution and quality-adjusted coverage, not prompt counts alone.
Is a process partly or fully agentic?	Process agenticity model based on steps, decisions, controls and human intervention.

Are agents performing like digital employees?	Agent master record, probation, scorecards, reviews, promotion, restriction and retirement.
How do we keep agents safe and reliable?	Harness engineering, evaluation, observability, approvals, access control and kill switch.
What value is being generated?	Baseline-to-outcome Value Realization with verified benefit, total cost and useful intelligence per AED.
What is the recommended technology approach?	Neptune-led portal with federated runtimes, common data layer, Superset analytics and Grafana observability.

2. CAIO Vision and Framework Principles

2.1 Vision

Create an AI-native enterprise where humans and agents work through explicit roles, approved procedures, engineered controls and measurable outcomes.

The framework should start with the approved organization and operating model — not with a list of technologies. AI opportunities should be derived from strategic objectives, section mandates, areas of activity, Quality Procedures, jobs and process steps. Each proposed agent must have a clear organizational home, business purpose, accountable owner, operating boundary and value target.

2.2 Design principles

- **Organization-led:** begin from division, super department, department, section, position and mandate.
- **Outcome-led:** every AI capability must improve a measurable business or strategic outcome.
- **Procedure-grounded:** approved Quality Procedures define the expected work sequence, controls and evidence.
- **Human accountable:** agents may execute work, but legal, ethical and managerial accountability remains human.
- **Harnessed by design:** no production agent without instructions, tools, guardrails, evaluations, logging and fallback.
- **Federated technology:** support SAP, Microsoft, Neptune and custom agents through a common control plane.
- **Privacy-respecting measurement:** measure work contribution and outcomes without turning prompt logs into employee surveillance.
- **Reuse before rebuild:** standardize reusable agent skills, harness templates, policies, APIs and evaluation suites.

- **Value and cost transparency:** link each agent run to outcomes, total cost, tokens and validated benefit.
- **Progressive autonomy:** move from copilot to managed agent to bounded autonomy only when evidence supports it.

2.3 AI-native maturity progression

Stage	Operating pattern	Typical outcome
1. Digital foundation	Systems, workflows, structured data and approved knowledge	Reliable digital execution and traceability
2. AI copilots	Humans use AI for drafting, searching, summarizing and analysis	Productivity and decision support
3. AI brains	Enterprise context, semantic knowledge and reusable skills	Consistent contextual intelligence
4. Managed agents	Agents perform defined tasks with human review or approval	Higher throughput and reduced manual effort
5. Autonomous agents	Agents execute end-to-end within approved boundaries	Digital operations with exception-based supervision
6. Adaptive enterprise	Agents and humans improve processes through governed learning	Continuous optimization and organizational agility

3. Enterprise Framework Overview

The framework consists of ten connected pillars. They should be implemented as one operating model even when different technology products provide individual capabilities.

Pillar	Purpose	Primary outputs
1. Strategy and Excellence	Connect AI to enterprise strategic objectives and excellence criteria.	AI Rooms, scorecards, strategic alignment.
2. Human-Agent Organization	Show human-only, human+agent and agent-only roles in the organization.	Digital workforce map, agent reporting lines.
3. Quality Procedures	Translate approved procedures into structured steps, controls and evidence.	QP registry, procedure compliance and AI opportunities.
4. Process Agenticity	Assess how much of each process is human, AI-assisted or agent-led.	Agenticity heatmap and target-state roadmap.

6. AI Playbook	Provide section- and division-level guidance on what to adopt, build and govern.	Living playbooks and rollout recommendations.
7. Data and Integration	Unify organization, process, agent, telemetry, cost and outcome data.	Common data model and secured APIs.
8. Agent Performance and Governance	Manage digital workers through lifecycle, scorecards and controls.	Agent reviews, incidents, promotion and retirement.
9. Value Realization	Validate baseline, outcomes, benefits, costs and capacity redeployment.	VR records and benefits assurance.
10. Token Economics and Observability	Understand consumption, reliability and cost per successful outcome.	Token ledger, traces, alerts and optimization.

Target outcome — The platform is not only an AI initiative register. It becomes the enterprise system of record for how humans and agents jointly perform work.

3.1 End-to-end conceptual model

Enterprise layer	Key entities
Direction	Strategic objectives, excellence criteria, policies and AI-native priorities
Organization	Division, super department, department, section, mandate and position
Work design	Job descriptions, areas of activity, Quality Procedures, processes and decisions
Workforce allocation	Human-only, human+copilot, human+managed agent, agent-only
Execution design	Agent, skill, harness, model, tool, API, knowledge source and approval
Evidence	Work item, agent run, Copilot activity, acceptance, override, quality and SLA
Outcomes	Capacity released, service improvement, compliance, financial and strategic value
Economics	Licenses, infrastructure, tokens, support, cost per outcome and net value

PART B

Human-Agent Organization & Process Design

Sections 4–5

4. Human-Agent Organization and Digital Workforce

4.1 Organization structure

DEWA → Division → Super Department → Department → Section → Position → Human / Human+Agent / Digital Employee

Agents should be visible in the operating model rather than remaining hidden inside applications. A digital employee may be displayed in the organization chart, provided its accountable human owner, manager, technical owner and risk owner are clearly identified.

4.2 Workforce patterns

Pattern	Definition	Illustrative example
Human-only	Human performs the activity; AI is not materially involved.	Final statutory approval or sensitive employee decision.
Human + Copilot	AI supports drafting, analysis, search or summarization; human owns the output.	Analyst prepares a management report using Copilot.
Human + Managed Agent	Agent performs defined work; human reviews exceptions or approves outcomes.	Procurement analyst supervises quotation comparison agent.
Agent-only within boundary	Agent executes approved work unattended with monitored exceptions.	Routine validation agent processes compliant low-risk cases.

4.3 Digital Employee Master Record

Category	Required attributes
Identity	Agent ID, digital job title, organization assignment, status and deployment environment.
Ownership	Business owner, agent manager, technical owner, risk owner and escalation authority.
Work assignment	Processes, activities, systems, data, decisions and permitted actions.

Authority	Autonomy level, financial limits, approval points and prohibited actions.
Performance	Outcome, quality, throughput, SLA, override, cost, reliability and compliance.
Lifecycle	Proposed, training, shadow, probation, active, restricted, suspended and retired.
Governance	Risk tier, impact assessment, evaluation status, last change and next review.

4.4 Measuring AI contribution to human work

Copilot usage counts do not establish how much meaningful work was completed. Measurement should start from job activities and baseline effort.

Coverage formula:

$$\text{AI Work Coverage \%} = \frac{\sum (\text{Activity baseline hours} \times \text{AI contribution \%})}{\sum \text{Activity baseline hours}}$$

Quality adjustment:

$$\text{Quality-Adjusted AI Coverage} = \text{AI Work Coverage} \times \text{Acceptance Rate} \times \text{Accuracy Rate}$$

Contribution code	Meaning
H100	Fully human.
H75/A25	Limited AI assistance.
H50/A50	Shared execution.
H25/A75	Agent performs most work; human validates.
A100-S	Autonomous execution with exception supervision.
A100-U	Unattended within approved authority.

5. Quality Procedures and Process Agenticity

5.1 Why Quality Procedures are essential

Quality Procedures are a key enterprise source because they describe the approved way a section performs and controls its work. They connect section mandates, roles, process steps, inputs, outputs, approvals, evidence, SLAs and exceptions. They are therefore valuable for identifying AI opportunities, designing harnesses, evaluating compliance and establishing Value Realization baselines.

Use of QP	Application in the framework
Source	Provides approved steps, rules, roles, controls, templates and evidence requirements.
Design input	Identifies which tasks are suitable for human-only, copilot, managed agent or autonomous execution.
Harness specification	Transforms procedure steps into instructions, tools, checks, approvals and escalation.
Evaluation standard	Determines whether the agent followed mandatory controls and produced required evidence.
Value baseline	Provides existing SLA, effort, rework, error and compliance indicators.
Audit evidence	Links agent logs and work items to the approved procedure version.

5.2 Structured Quality Procedure model

QP master	Procedure-step attributes
QP ID, title and version	Step ID and activity description
Effective and review dates	Responsible and accountable roles
Owning organization unit	Inputs, outputs and system
Process and policy links	Business rules and controls
Official document link	Evidence and record requirements
Status and approval	Approval, exception, SLA and KPI
Quality owner	Current and target execution mode
Risk classification	Assigned agent, harness and human checkpoint

5.3 Process agency levels

Level	Classification	Description
L0	Manual	Human execution without meaningful automation.
L1	Digitized	Workflow and systems support, but decisions remain human.
L2	AI-assisted	AI drafts, summarizes, searches or recommends.
L3	Partly agentic	Agents execute selected steps under managed supervision.

L5	Autonomous within boundary	End-to-end execution within defined authority and controls.
L6	Self-optimizing under governance	Controlled adaptive improvement; restricted for high-risk utility processes.

Control principle — Do not automate an unclear procedure. Validate, simplify and approve the Quality Procedure before converting it into an agent workflow.

PART C

Harness Engineering & AI Playbook

Sections 6–7

6. Harness Engineering

6.1 Definition

Harness Engineering is the discipline of designing the operational environment that makes an AI model or agent reliable, secure, measurable and fit for enterprise work. The model provides intelligence; the agent provides goals and actions; the harness governs how the agent receives context, uses tools, follows procedures, makes decisions, requests approvals, handles failure and proves what it did.

Harness equation

Production Agent = Model + Instructions + Context + Tools + Workflow + Permissions + Guardrails + Evaluations + Observability + Human Controls + Cost Limits

6.2 Harness components

Component	Purpose
Role and instructions	Define the agent purpose, boundaries, tone, prohibited actions and expected outputs.
Context and knowledge	Provide approved QPs, policies, JDs, process data, enterprise standards and work-item context.
Tools and APIs	Control access to SAP, D2D, documents, databases and other systems.
Workflow orchestration	Define sequence, branching, state, retries and dependencies.
Identity and permissions	Apply least privilege, service identity, segregation of duties and transaction limits.
Guardrails	Validate input/output, sensitive data, policy compliance and permitted actions.
Human checkpoints	Require review, approval or escalation for defined decisions and risk conditions.
Evaluation suite	Test quality, accuracy, procedure compliance, safety, consistency and adversarial cases.

Cost controls	Set model routing, token budgets, retry limits and automatic suspension thresholds.
Fallback and kill switch	Define safe failure, manual takeover, rollback, suspension and retirement.
Version control	Track instruction, model, tool, QP, knowledge and evaluation changes.

6.3 Harness lifecycle

Stage	Required gate
Design	Approved digital JD, process, QP, owner, risk tier and success criteria.
Build	Configured instructions, tools, context, controls and telemetry.
Evaluate	Representative, edge, adversarial and unauthorized test cases passed.
Shadow mode	Agent runs without executing final actions; results compared with humans.
Probation	Restricted production scope and mandatory human approval.
Active	Approved autonomy, monitoring, budgets and performance review.
Change	Re-evaluation after model, prompt, tool, data or QP change.
Restrict/Retire	Suspend for incident, poor performance, obsolescence or policy change.

6.4 D2D BRD Agent example

Harness area	Illustrative design
Context	Demand record, division, process, similar demands, EA standards and approved templates.
Tools	Read D2D, retrieve knowledge, identify duplicates, generate sections, save draft and route for approval.
Guardrails	Do not invent facts, expose restricted data, change approved fields or submit without human confirmation.
Evaluations	Completeness, accuracy, traceability, stakeholder identification, compliance and human acceptance.
Observability	Sources retrieved, tools called, tokens, edits, overrides, errors, time and final outcome.

7. AI Transformation Playbook

The playbook should be a living, data-driven product rather than a static PDF. It should combine enterprise standards with division and section-specific recommendations. The approach is consistent with external AI transformation playbooks that organize adoption through copilots, AI brains and autonomous agents, and that map agents to organizational functions and phased rollout.

Playbook module	Content
1. AI-native vision	Strategic purpose, principles, target operating model and non-negotiable controls.
2. Organization design	Current and target human-agent organization, digital reporting lines and workforce transition.
3. Copilot productivity	Approved day-one uses for documentation, search, analysis, meetings and development.
4. Department playbooks	Mandate, processes, QPs, pain points, agents, harnesses, readiness, risks and roadmap.
5. AI brain and knowledge	Enterprise terminology, policies, procedures, historical cases and knowledge ownership.
6. Agent catalogue	Reusable enterprise agents, division agents, skills, dependencies and approved models.
7. Buy/configure/build guide	SAP, Microsoft, Neptune, custom and hybrid decision criteria.
8. Responsible AI	Risk tiers, human oversight, security, privacy, audit and incident response.
9. Value Realization	Baselines, targets, outcomes, benefits, validation and post-go-live review.
10. Token economics	Budgets, cost per outcome, routing, retries, caching and optimization.
11. Maturity and rollout	Copilot → AI brain → managed agents → bounded autonomy → continuous improvement.

7.1 AI Rooms

AI Rooms should organize transformation around strategic domains, not merely technologies. Each AI Room should include an executive sponsor, relevant divisions, strategic outcomes, priority processes, human-agent map, experiment backlog, active agents, performance, risk, value and playbook updates.

- Customer Experience AI Room
- Asset and Maintenance AI Room

- Corporate Services AI Room
- Sustainability AI Room
- Project Delivery AI Room
- Knowledge and Productivity AI Room
- Revenue and Billing AI Room

PART D

Data, Value & Token Economics

Sections 8–9

8. Data, Integration and Measurement Model

8.1 Common enterprise data model

A common model is required before connecting individual tools. The platform must support navigation from strategy to organization, procedure, process, agent, run, outcome and value.

Domain	Core entities
Strategy and excellence	Strategic objective, excellence criterion, KPI, AI Room.
Organization	Org unit, position, employee, mandate, JD, area of activity.
Quality and process	Quality Procedure, procedure step, process, process step, decision, control.
Digital workforce	Agent, skill, harness, autonomy level, owner and performance record.
Execution	Work item, agent run, Copilot activity, tool call, human approval, exception.
Technology	System, API, model, knowledge source, data product and environment.
Economics	Token usage, license, infrastructure, implementation, support and total cost.
Outcomes	Operational metric, business outcome, benefit, VR record and validation.

8.2 Copilot integration

Microsoft reporting can provide adoption and engagement indicators for Microsoft 365 Copilot and organizationally built agents. The framework should use this telemetry for adoption, active use, application use and licensing insights. It should not treat usage counts as a direct measure of productivity or job coverage.

Source	Use	Limitation / control
Microsoft Copilot usage reports/APIs	Enabled users, active users, active days, app engagement and usage trends.	Availability and granularity depend on licensing, report scope and privacy controls.

D2D and work systems	Work item, process step, AI output, acceptance, cycle time and outcome.	Requires standardized work-item and agent identifiers.
Harness telemetry	Tools, tokens, latency, errors, quality, overrides and compliance.	Must be standardized across different runtimes.
Employee/activity assessment	Baseline hours and contribution allocation.	Should be validated periodically and not used as surveillance.

8.3 Work Contribution Ledger

The Work Contribution Ledger should correlate technical usage with work evidence. Recommended fields include employee or pseudonymous ID, organization, role, process, activity, work item, agent, timestamp, baseline effort, actual human effort, AI contribution, acceptance, correction, quality, outcome, tokens and cost.

Measurement rule — Observed usage ≠ verified work contribution ≠ realized business value.
These three measures must remain separate in all dashboards.

9. Value Realization and Token Economics

9.1 Value hierarchy

Level	Examples
Usage	Active users, agents, sessions, prompts, tokens and transactions.
Operational performance	Cycle time, SLA, throughput, accuracy, rework, exception rate and availability.
Business outcome	Cost avoided, revenue protected, service quality, asset performance, compliance and risk reduction.
Strategic value	Strategic objective, excellence improvement, resilience, new capability and workforce redeployment.

9.2 Core formulas

Primary AI value principle:

Useful Intelligence per AED =
Successful Business Outcomes ÷ Total AI Cost

Net value:

Cost per successful transaction:

$$\text{Cost per Successful Transaction} = \frac{\text{Total Agent Cost}}{\text{Successful Transactions}}$$

Cost per verified hour released:

$$\text{Cost per Verified Hour Released} = \frac{\text{Total AI Cost}}{\text{Verified Human Hours Released}}$$

Total AI Cost should include model and token cost, licenses, infrastructure, data and integration, implementation, testing, assurance, human supervision, maintenance, support, security, compliance and change management.

9.3 Token economics hierarchy

Enterprise → Division → Department → Process → Agent → Harness/Skill → Model → Transaction

Token KPI	Management use
Tokens per successful outcome	Compare the efficiency of agents and harness versions.
Cost per accepted output	Identify expensive or low-acceptance use cases.
Retry-token ratio	Detect poor orchestration, unclear instructions or unstable tools.
Wasted-token ratio	Measure failed, abandoned or rejected consumption.
Small-model routing rate	Track economical routing of low-complexity tasks.
Cache utilization	Measure reuse of stable context and reduced cost.
Human correction per 1,000 tokens	Connect consumption to quality and editing burden.
Value per million tokens	Executive comparison of AI economics across domains.

9.4 VR workflow

Stage	Action
Baseline	Capture pre-AI effort, cost, quality, SLA, volume and risk.
Target	Approve expected outcome, benefit, measurement method and owner.
Implementation	Link initiative, process, agent and harness.

Evidence	Automatically collect operational performance and cost.
Validation	Business, Finance/BPI/PMO validate realized benefit.
Review	Compare forecast versus actual and decide scale, optimize or retire.

PART E

Technology & Governance

Sections 10–11

10. Technology and Visualization Architecture

10.1 Recommended hybrid architecture

Unified Experience	SAP Neptune portal: organization, QPs, processes, agents, playbooks, approvals and VR forms
Specialized Visualization	Custom interactive organization graph · Superset executive/VR analytics · Grafana observability
Enterprise Data & Control	Organization master, process/QP model, agent/harness registry, work ledger, token ledger and VR
Harness & Orchestration	Instructions, tools, approvals, evaluations, guardrails, routing, fallback and telemetry
Federated AI Runtimes	Microsoft Copilot/agents · SAP/Joule capabilities · Neptune agents · custom agents/models
Enterprise Systems & Knowledge	D2D, SAP, HR, Finance, SharePoint/Documentum, data platforms and approved knowledge

10.2 Recommended platform roles

Capability	Recommended platform role
Business portal and workflow	SAP Neptune for integrated user experience, data maintenance, approvals and process interaction.
Organization visualization	Custom graph embedded in Neptune for large-scale expand/collapse, filters, relationships and heatmaps.
D2D outcomes	D2D APIs as source for demand, stage, AI usage, SLA, approval and work-item outcomes.
Copilot adoption	Microsoft reporting/APIs for authorized usage and agent adoption indicators.
Executive analytics and VR	Apache Superset embedded in Neptune.

Master and transactional data	Enterprise relational data store exposed through governed APIs.
Telemetry/event data	Scalable event and observability store.
Knowledge	Approved enterprise repositories with ownership, versioning and access control.
Agent execution	Federated runtimes; do not force all agents into a single product.

10.3 Technology option assessment

Option	Strengths	Recommended role
SAP Neptune	Rapid enterprise app delivery, SAP alignment, API-driven integration and embedded custom UI.	Primary business portal and workflow layer.
Microsoft stack	Strong Copilot integration, identity, reporting and low-code agent development.	Use where Microsoft 365/Copilot is the work environment and approvals permit.
SAP/Joule/BTP	Strong SAP process context and ERP-domain capabilities.	Use for SAP-centric agents and skills, subject to hosting and residency assessment.
Fully custom	Maximum control, interoperability and tailored organization graph/control plane.	Use selectively for strategic common services, complex visualization and cross-platform control.
Hybrid	Combines rapid delivery and best-fit specialist components.	Recommended target architecture.

11. Governance, Risk and Performance

11.1 Governance bodies

Body	Primary responsibilities
Enterprise AI Council	Strategy, standards, investment, risk appetite, cross-enterprise priorities and escalation.
AI Workforce Office	Human-agent org design, digital JDs, agent registry, workforce impact, skills and lifecycle.
Agent Review Board	Production approval, autonomy, model/tool changes, incidents, restrictions and retirement.

	maintenance.
Value Validation Forum	Baseline integrity, realized benefits, cost validation and capacity redeployment.
Harness Engineering Practice	Reusable harness patterns, evaluation libraries, observability standards and release gates.

11.2 Mandatory ownership roles

Role	Accountability
Business owner	Outcome, policy, scope and continued business need.
Agent manager	Daily performance, exceptions, user feedback and operational review.
Technical owner	Build, integration, availability, change and support.
Risk owner	Risk assessment, controls, compliance, incidents and independent challenge.
Data/knowledge owner	Accuracy, access, versioning and refresh of knowledge sources.
Value owner	Baseline, benefit measurement, validation and realization actions.

11.3 Agent performance scorecard

Dimension	Suggested weight	Illustrative KPI
Business outcome	25%	Successful process outcomes and strategic KPI contribution.
Quality	20%	Accuracy, acceptance and first-time-right.
Productivity	15%	Throughput and cycle-time improvement.
Reliability	10%	Successful run rate, availability and recovery.
Human collaboration	10%	Override, escalation and supervisor satisfaction.
Cost efficiency	10%	Cost per outcome and token efficiency.
Compliance and safety	10%	Procedure compliance and zero critical violations.

Performance principle — Compliance is a gating condition. An agent that is fast and inexpensive but breaches policy or access rules should fail its performance review.

11.4 Employee performance in an AI-native model

Employees should be assessed on effective and responsible AI leverage, not on prompt volume. Relevant measures include output quality, productive redeployment of released capacity, agent supervision, exception handling, process improvement, reusable knowledge contribution, strategic outcomes and responsible use.

PART F

Roadmap & Execution

Sections 12–14

12. Implementation Roadmap

Phase	Indicative duration	Key deliverables
0. Mobilize	4–6 weeks	Vision, principles, governance, terminology, data model, privacy approach, priority divisions and architecture.
1. Enterprise foundation	8–12 weeks	Org hierarchy, JDs, QPs, processes, AI initiatives, agent registry, playbook, initial graph and VR/token foundations.
2. D2D pilot	12–16 weeks	D2D harness, work ledger, process agenticity, AI coverage, outcome evidence and first validated VR case.
3. Copilot and telemetry integration	3–6 months	Authorized Microsoft usage integration, agent telemetry, dashboards, performance and token budgets.
4. Value Realization scale-out	3–6 months	Automated baselines/outcomes, validation workflow, capacity redeployment and executive scorecards.
5. AI Rooms enterprise scale	6–18 months	Division playbooks, reusable harnesses, agent catalogue, federated rollout and maturity reviews.

12.1 Delivery workstreams

- **Business architecture:** organization, mandates, jobs, areas of activity, QPs and process model.
- **Digital workforce:** agent registry, digital JDs, ownership, lifecycle and performance.
- **Harness engineering:** patterns, evaluations, guardrails, orchestration and observability.
- **Data and integration:** common model, APIs, work ledger, telemetry and knowledge.
- **Platform:** Neptune application, custom graph, embedded dashboards and user experience.
- **Value and finance:** baselines, cost model, tokens, benefits and validation.
- **Governance and change:** policy, privacy, employee engagement, training and adoption.

12.2 Build sequence

Release	Scope
Release 1	Organization baseline, human-agent classification, QP registry, process links and agent master.
Release 2	D2D integration, BRD harness, work evidence, AI contribution and initial observability.
Release 3	Copilot integration, adoption dashboards, role-level indicators and license insights.
Release 4	VR module, token ledger, total cost, Finance/BPI validation and executive analytics.
Release 5	Harness templates, evaluation libraries, model routing, incident management and enterprise scale.

13. Recommended MVP and D2D Pilot

D2D is the recommended first end-to-end implementation because it already combines organizational users, process stages, AI BRD generation, classification, stakeholder identification, SLAs, human verification, Value Realization and token-cost considerations.

13.1 MVP scope

- Complete organization hierarchy and selected section profiles.
- 20–30 representative JDs and areas of activity.
- 5–10 critical Quality Procedures.
- 10–15 priority processes with step-level agenticity.
- Existing AI initiative and agent inventory.
- Agent and harness master records.
- D2D integration and Work Contribution Ledger.
- Basic AI coverage and agent performance.
- VR and token ledger foundation.
- One AI Room and one executive dashboard.

13.2 D2D pilot data flow

Step	Flow
1	Business submits demand and D2D assigns Demand ID.
2	Copilot/agent supports intake, classification, stakeholder identification and BRD preparation.

4	D2D records accepted output, stage, SLA, approval and final outcome.
5	Neptune correlates role, section, QP, process, agent and work contribution.
6	Superset updates strategic, AI coverage and VR dashboards.
7	Grafana updates harness performance, errors, latency, token consumption and alerts.
8	Business and Finance/BPI validate realized benefit and next-scale decision.

13.3 Pilot success criteria

Dimension	Pilot target approach
Coverage	Demonstrate organization-to-QP-to-process-to-agent traceability.
Quality	Measure acceptance, correction, completeness and procedure compliance.
Productivity	Validate baseline and actual processing effort and cycle time.
Governance	Complete owner, risk, evaluation, approval and audit evidence.
Economics	Capture total cost, tokens, cost per outcome and verified value.
Scalability	Reuse at least one harness template or agent skill in a second process.

14. Executive Dashboards and KPIs

Dashboard	Key views
AI-Native Organization Map	Human-only, human+agent, digital employees, reporting lines, AI coverage, risk and value.
Process Agenticity Heatmap	Current/target level by division, section, QP and process; readiness and risk.
Workforce AI Coverage	Observed adoption, verified contribution, quality-adjusted coverage, hours released and redeployment.
Digital Employee Performance	Outcome, quality, SLA, reliability, overrides, incidents, cost and review status.
Harness Observatory	Run traces, latency, tool errors, evaluations, version comparison, tokens and alerts.
Strategic Alignment	AI initiatives and agents mapped to objectives, excellence and enterprise KPIs.

intelligence per AED.

Token Economics	Cost by division/process/agent, tokens per outcome, retry waste, model mix and budget variance.
Playbook Progress	Sections assessed, QPs structured, processes mapped, agents active, controls complete and lessons updated.

14.1 Core executive indicators

Indicator	Definition
AI adoption	Percentage of eligible users or roles using approved AI capabilities.
Verified AI work contribution	Activity-weighted share of work supported or performed by AI.
Quality-adjusted coverage	AI contribution adjusted for acceptance and accuracy.
Process agency	Share of process steps and decisions performed by agents, adjusted for risk and intervention.
Procedure compliance	Applicable mandatory QP controls completed and evidenced.
Agent performance index	Weighted outcome, quality, productivity, reliability, collaboration, cost and compliance.
Verified hours released	Baseline human effort less actual human effort, validated by work evidence.
Productive redeployment	Released capacity applied to approved higher-value work.
Useful intelligence per AED	Successful business outcomes divided by total AI cost.
Token efficiency	Tokens or cost per successful and accepted business outcome.

PART G

Decisions & Appendices

Section 15 & Appendices A–D

15. Key Decisions and Success Factors

15.1 Decisions required from CAIO and governance

- Official terminology for digital employee, enterprise agent and managed agent.
- Whether agents appear formally in organization views and departmental scorecards.
- Permitted employee-level telemetry, privacy controls and transparency requirements.
- Decisions and processes that must always retain human authority.
- Agent autonomy approval levels and promotion criteria.
- Standard harness, evaluation, observability and incident requirements.
- Enterprise platform principles and approved deployment environments.
- Token budget ownership and total AI cost allocation.
- Value baseline and benefit validation authority.
- Approach to redeploying verified capacity released by AI.

15.2 Critical success factors

Success factor	Practical implication
Reliable source data	Clean organization, JD, QP, process and ownership data before advanced AI mapping.
Clear accountability	No agent without a business owner, agent manager, technical owner and risk owner.
Procedure readiness	Outdated or ambiguous QPs must be revised before automation.
Evidence-based measurement	Use work-item and outcome data rather than self-reported savings or prompt counts.
Reusable harness engineering	Build standard patterns instead of one-off agent wrappers.
Progressive autonomy	Use shadow and probation modes before higher autonomy.

Technology neutrality	Keep master data, telemetry and control standards portable across vendors.
Value governance	Stop or redesign agents that do not produce validated value.
Workforce engagement	Explain how roles evolve and avoid framing AI telemetry as surveillance.

Executive recommendation — Recommended decision: proceed with a Neptune-led MVP using D2D as the first use case, while establishing the common data model and Harness Engineering standards from day one.

A. Appendix A · Reference Data Model

Entity	Selected relationships
ORG_UNIT	contains positions, owns QPs/processes, sponsors initiatives and receives value.
POSITION	belongs to org unit, has JD, performs activities and is assigned human/agent support.
EMPLOYEE	occupies position, supervises agents and performs/approves work.
QUALITY_PROCEDURE	owned by section, decomposes into steps, controls and evidence.
PROCESS / PROCESS_STEP	linked to QP, roles, systems, decisions, agenticity and outcomes.
AI_INITIATIVE	derived from strategy/process/QP and implemented through agents/harnesses.
AGENT	assigned to org/process/activity and governed by owners, status and scorecard.
HARNESS	defines instructions, tools, knowledge, controls, evaluations and version.
WORK_ITEM	business case/demand/transaction supported by human and agent activity.
AGENT_RUN	execution trace linked to harness, model, tools, tokens, quality and outcome.
VALUE_REALIZATION	links baseline, target, actual outcome, cost, validation and strategic benefit.
TOKEN_USAGE	allocated to division, process, agent, harness, model and transaction.

B. Appendix B · Illustrative Agent and Harness Records

B.1 Digital employee record

Field	Illustrative entry
Agent ID	AGT-D2D-001
Digital job title	D2D Intake and BRD Agent
Organization	I&TF / Demand-to-Delivery
Business owner	D2D Product Owner
Agent manager	D2D Governance Lead
Technical owner	Assigned platform team
Risk owner	AI Governance / Cybersecurity
Assigned work	Demand completeness, classification, duplicate check, stakeholder suggestion and BRD draft
Autonomy	Managed agent; human approval required
Status	Probation / active after review
Primary KPIs	Acceptance, completeness, SLA, procedure compliance, cost per accepted BRD

B.2 Harness record

Field	Illustrative entry
Harness ID	HRN-D2D-BRD-001
Agent	AGT-D2D-001
QP/process	Demand intake, shaping and BRD preparation
Instructions	Approved D2D rules, role boundaries and output format
Knowledge	D2D policy, BRD templates, EA standards and prior approved examples
Tools	D2D APIs, search, document generation and workflow routing
Human checkpoints	Business confirmation, IT Lead review and management approval
Evaluations	Accuracy, completeness, traceability, stakeholder fit, compliance and adversarial tests
Token controls	Per-run ceiling, retry limit, approved model routing and alerts

Observability	Trace, tool call, source, token, latency, quality, override and outcome
Version triggers	Model, instruction, QP, API, knowledge or control change

C. Appendix C · CAIO Whiteboard Reference

The following images were provided as the source discussion for the organization-led AI-native framework. The consolidated document interprets the key themes as organization structure, people and agents, process and technology, AI Rooms, excellence, benchmarking, playbooks, harness engineering, Quality Procedures, strategic outcomes, Value Realization and token economics.

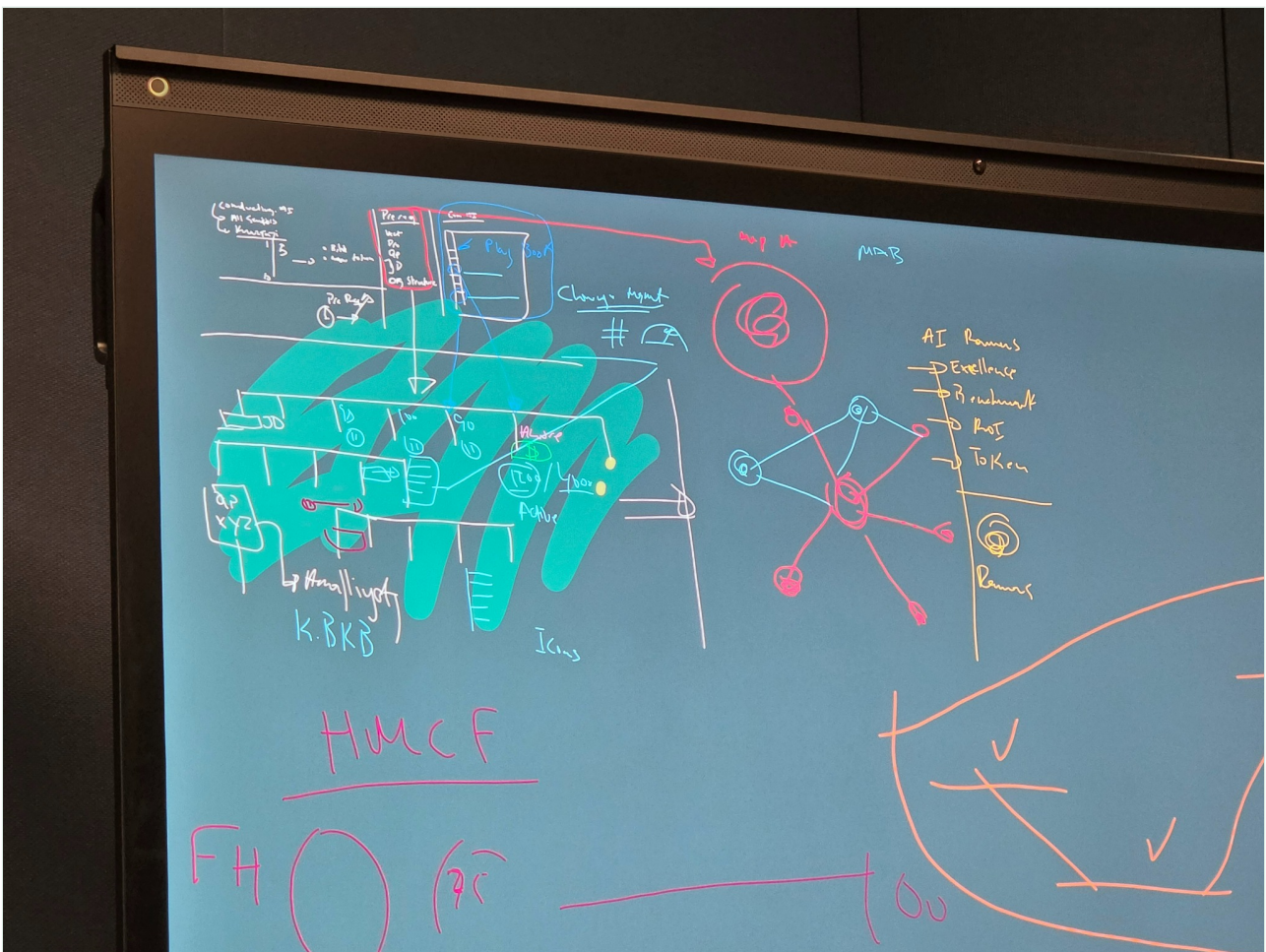


Figure C1. CAIO direction whiteboard reference

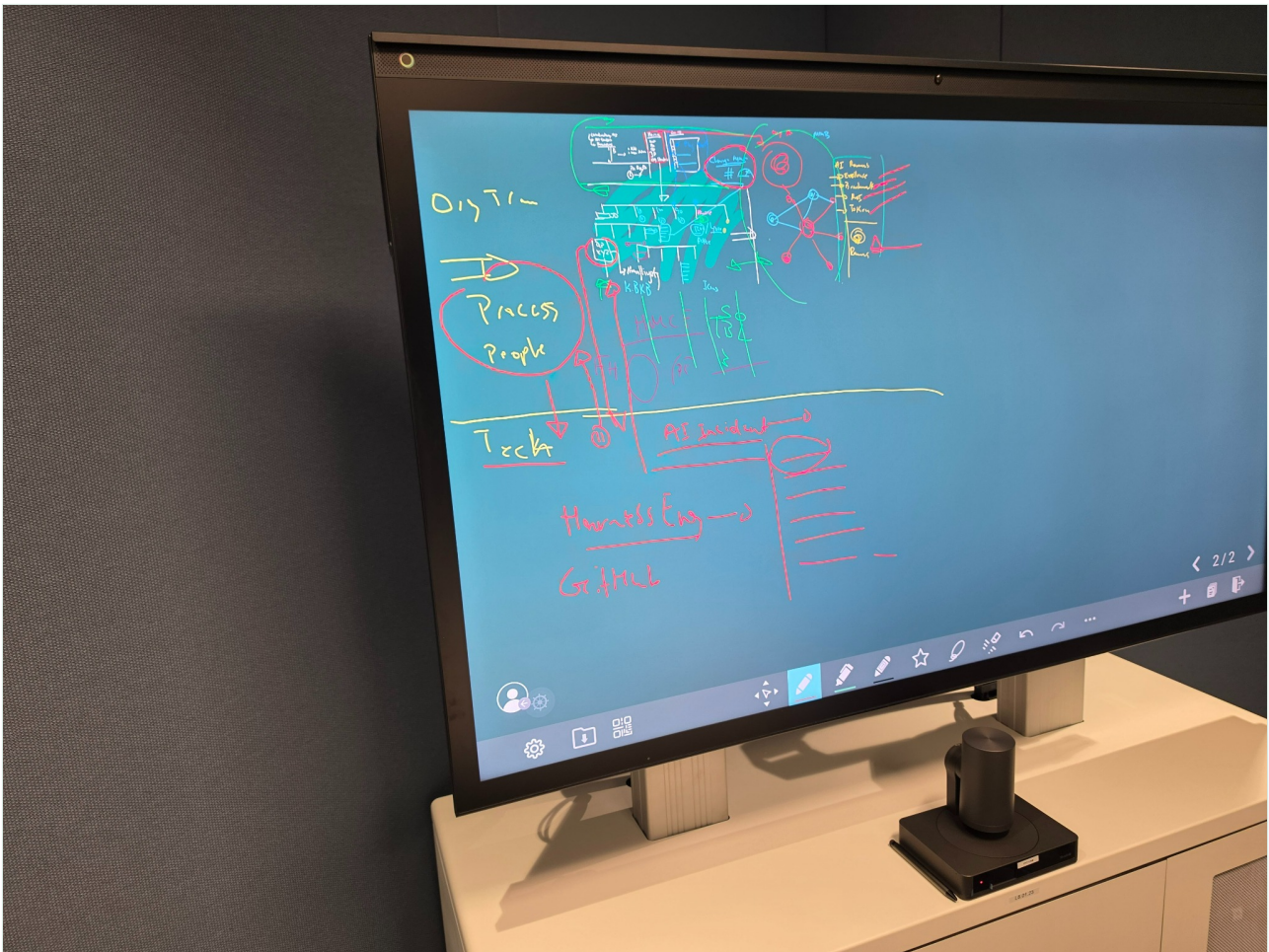


Figure C2. CAIO direction whiteboard reference

D. Appendix D · External Reference Sources

These sources informed selected technology and market-reference observations. Product capabilities should be revalidated during detailed solution design and procurement.

#	Source
1	Conducting AI — AI-native company designs and transformation playbook: https://www.conducting.ai/ai-native-designs
2	Conducting AI — Home and three-phase playbook overview: https://www.conducting.ai/
3	Microsoft 365 Copilot usage report: https://learn.microsoft.com/en-us/microsoft-365/admin/activity-reports/microsoft-365-copilot-usage
4	Microsoft 365 Copilot agent usage report: https://learn.microsoft.com/en-us/microsoft-365/admin/activity-reports/microsoft-365-copilot-agents

- 5 **Microsoft 365 Copilot APIs overview:**
<https://learn.microsoft.com/en-us/microsoft-365/copilot/extensibility/copilot-apis-overview>
- 6 **Microsoft Copilot Control System — measurement and reporting:**
<https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-control-system/measurement-reporting>
- 7 **Neptune DXP — API integration:**
<https://docs.neptune-software.com/neptune-dxp-open-edition/24.14/addon-guide/core-api-integration.html>
- 8 **Neptune Naia Build — overview and boundaries:**
<https://docs.neptune-software.com/naia-build/overview/naia-build.html>
- 9 **Grafana node graph visualization:**
<https://grafana.com/docs/grafana/latest/visualizations/panels-visualizations/visualizations/node-graph/>
- 10 **Apache Superset embedded dashboards:**
<https://superset.apache.org/docs/embedded-superset/>

— End of Document —